

```

Stack Menu
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice:1
Enter data:1

Stack Menu
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice:1
Enter data:2

Stack Menu
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice:1
Enter data:3

Stack Menu
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice:1
Enter data:4

Stack Menu
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice:1
Enter data:5

Stack Menu
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice:1
Enter data:6
Overflow

Stack Menu
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice:4

Stack elements are:5
4
3
2
1

Stack Menu
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice:3
Top element:5

Stack Menu
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice:2
Popped:5

Stack Menu
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice:2
Popped:4

Stack Menu
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice:2
Popped:3

Stack Menu
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice:2
Popped:2

Stack Menu
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice:2
Popped:1

Stack Menu
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice:2
Underflow

Stack Menu
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice:5

Process returned 0 (0x0)   execution time : 223.051 s
Press any key to continue.

```

- Q) Write a program to simulate the working of stack using an array with the following:
 a) Push b) Pop c) Display. The program should print appropriate messages for stack overflow, stack underflow.

```
#include <stdio.h>
```

```
#define N5
```

```
int stack[N];
```

```
int top = -1;
```

```
void push()
```

```
{
    int x;
```

```
    printf("Enter data: ");
```

```
    scanf("%d", &x);
```

```
    if (top == N-1)
```

```
    {
        printf("Overflow\n");
```

```
    }
    else
```

```
    {
        top++;
```

```
        stack[top] = x;
```

```
    }
```

```
}
```

```
void pop()
```

```
{
```

```
    int item;
```

```
    if (top == -1)
```

```
    {
```

```

    printf("UnderFlow\n");
}
else
{
    item = stack[top];
    top--;
    printf("Popped: %d\n", item);
}
}

```

```

void peek()
{
    if (top == -1)
    {
        printf("UnderFlow\n");
    }
    else
    {
        printf("Top element: %d\n", stack[top]);
    }
}

```

```

int main()
{
    int choice;
    while(1)
    {
        printf("\nStack Menu\n");
        printf("1. Push\n");
        printf("2. Pop\n");
        printf("3. Peek\n");
        printf("4. Exit\n");
    }
}

```

```
printf("Enter your choice:");
scanf("%d", &choice);
```

```
switch(choice)
{
```

```
case 1:
```

```
    push();
    break;
```

```
case 2:
```

```
    pop();
    break;
```

```
case 3:
```

```
    peek();
    break;
```

```
case 4:
```

```
    return 0;
```

```
default:
```

```
    printf("Invalid choice\n");
```

```
}
```

```
}
```

```
return 0;
```

```
}
```

O/P Stack Menu

1. Push

2. Pop

3. Peek

4. Exit

Enter your choice: 1

Enter data: 10

Stack Menu

1. Push

2. Pop

3. Peek

4. Exit

Enter your choice: 3

Top element: 10

Stack Menu

1. Push

2. Pop

3. Peek

4. Exit

Enter your choice: 2

Popped: 10

Stack Menu

1. Push

2. Pop

3. Peek

4. Exit

Enter your choice: 4

22/9/23

